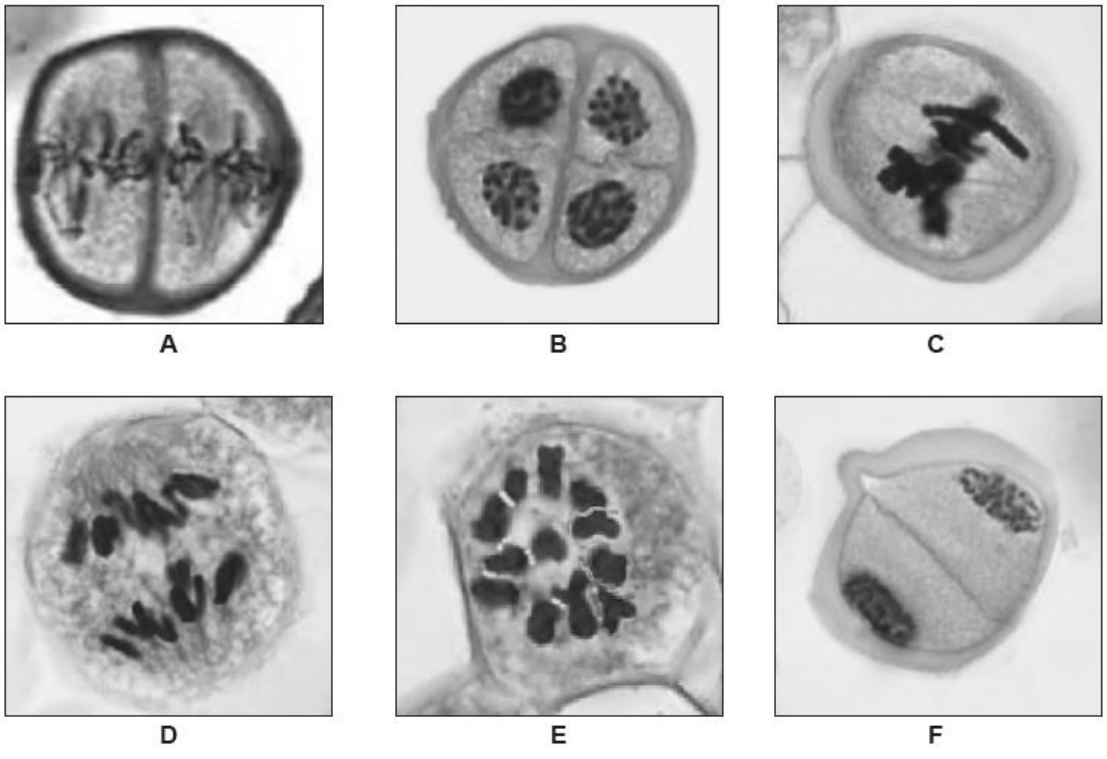
Structured questions

|!|BCELA0120000001|!| ★★ **Cross topic**

WJEC GCE A level Biology/Human Biology Module BY5 Jun 2016 Q2

The photomicrographs show some stages of cell division in a flower of a lily, *Lilium candidum*.



(a) (i) Identify the type of cell division shown. (1 mark)

_____

(ii) State a part of the flower that could have been used to produce the photomicrographs. (1 mark)

_____

(iii) Write the letters in the order the stages would occur (the first one has been done for you). (2 mark)

E → _____ → _____ → _____ → _____ → _____

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(iv) Use photomicrograph E to find the haploid number of *Lilium candidum*. (1 mark)

(b) (i) Name the stages shown in the following photomicrographs: A and C (1 mark)

(ii) Use your knowledge of cell division to describe ***two*** differences between the arrangement of chromosomes in stages A and C. (2 marks)

(c) Name ***two*** processes occurring during this type of cell division which help to ensure genetic variation in the offspring. (2 marks)

##

(a) (i) Meiotic cell division (1)

(ii) Anther / ovule (1)

(iii) E → C → D → F → A → B (2)

(iv) 6 (1)

(b) (i) A: Metaphase II; C: Metaphase I (1)

(ii) Any ***two*** of the following comparisons: (2)

C (metaphase I)	A (metaphase II)
Pairing of chromosomes	Chromosomes not paired
Centromeres not on equator	Centromeres on equator
Chromosomes either side of equator	Chromosomes on equator
Chiasmata present	No chiasmata

(c) Crossing over (1)

Independent assortment of homologous chromosomes (1)

(10 marks)

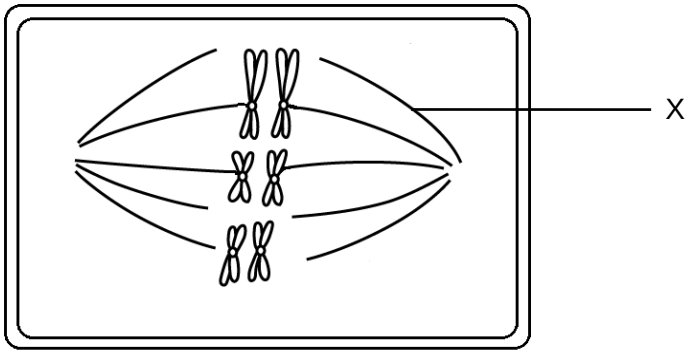
##

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	Skill assessed	Mark(s)
(a)(i)	Analysing data	1
(a)(ii)	Understanding basic concepts	1
(a)(iii)	Analysing data	2
(a)(iv)	Analysing data	1
(b)(i)	Analysing data	1
(b)(ii)	Understanding basic concepts	2
(c)	Understanding basic concepts	2

|!|BCELB0120000002|!| ★

The diagram below shows a plant cell carrying out cell division.



(a) Name this type of cell division. Explain your answer. (2 marks)

(b) Suggest a part of the plant where this type of cell division takes place. (1 mark)

(c) Name structure X and state its role in cell division. (2 marks)

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(d) Draw a simple diagram to show one of the daughter cells resulting from this cell division. (2 marks)

(e) State ***two*** processes that occur in the cell during interphase to prepare for this cell division. (2 marks)

_____

_____

_____

##

(a) Meiotic cell division (1)

The homologous pairs of chromosomes line up along the equatorial plane of the cell. (1)

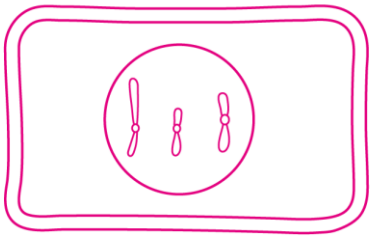
(b) Anther / ovule (1)

(c) Spindle fibre (1)

It shortens and pulls the two chromosomes of each homologous pair / chromatids of each chromosome apart to opposite poles of the cell. (1)

(d) Correct drawing: (2)

- Outline of the cell (1)
- Correct number and appearance of chromosomes (1)



(e) Any ***two*** of the following: (2)

- Replication of DNA (1)
- Synthesis of new organelles (1)

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- Synthesis of proteins (1)

(9 marks)

##

	Skill assessed	Mark(s)
(a)	Analysing data	2
(b)	Understanding basic concepts	1
(c)	Analysing data	1
	Understanding basic concepts	1
(d)	Presenting information	2
(e)	Understanding basic concepts	2

|!|BCELB0120000003|!| ★

The diagram below shows a plant cell undergoing meiosis.



(a) Identity the stage of meiosis shown in the diagram. (1 mark)

(b) State the main events that will occur to complete meiosis from this stage.

(3 marks)

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(c) Describe how meiosis could give rise to genetic variation. (5 marks)

##

(a) Metaphase II (1)

(b) The spindle fibres shorten and move the sister chromatids of each chromosome towards the opposite poles of the cell. (1)

A nuclear membrane forms around each group of chromosomes. (1)

The chromosomes loosen to become chromatin again. (1)

(c) Independent assortment of homologous chromosomes (1)

During metaphase I, the arrangement of homologous pairs along the equatorial plane is random. (1)

One chromosome from each pair can move along with either chromosome of another pair into a daughter cell. (1)

Crossing over (1)

The two non-sister chromatids of a homologous pair cross each other at certain points (chiasmata) and exchange corresponding segments. (1)

(9 marks)

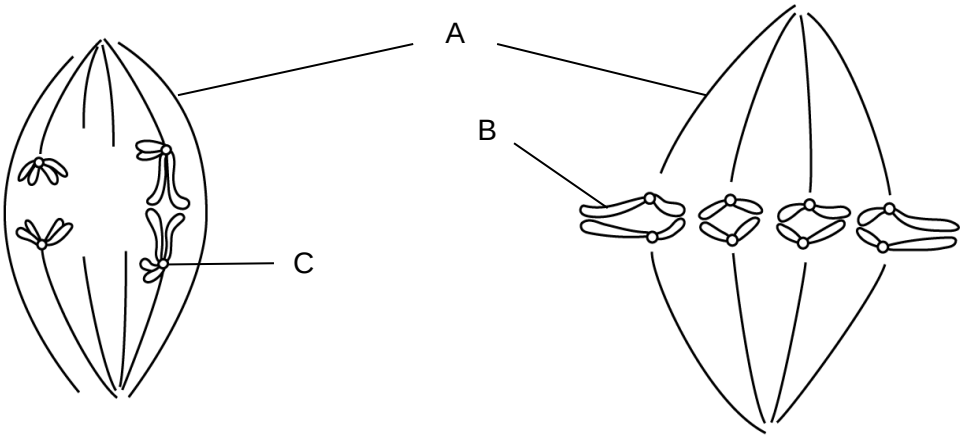
##

	Skill assessed	Mark(s)
(a)	Analysing data	1
(b)	Understanding basic concepts	3
(c)	Understanding basic concepts	5

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|!|BCELB0120000004|!| ★

The diagram below shows the process of cell division carried out by two cells in the same flowering plants. The chromosome number of this plant species is four.



Cell X Cell Y

(a) Identify the structures labelled A, B and C. (3 marks)

(b) State the stage of division in each cell. (2 marks)

(c) Cells X and Y are carrying out two different types of cell division. State the significance of the two types of cell division to flowering plants. (5 marks)

##

(a) A: Spindle fibre (1)
B: Chromatid (1)
C: Centromere (1)

(b) Cell X: Anaphase I / metaphase I (1)
Cell Y: Metaphase (1)

(c) Cell X (meiotic cell division):
To form haploid gametes. (1)
During fertilization, the diploid number of chromosome can be restored and the diploid number of the species can be kept constant generation after generation. (1)
This type of cell division also gives rise to genetic variation, which increase the survival change of the species when facing environmental changes. (1)
Cell Y (mitotic cell division):
To produce new cells that are genetically identical to the parent cell (1)
for the plant to grow / repair damaged tissues. (1)

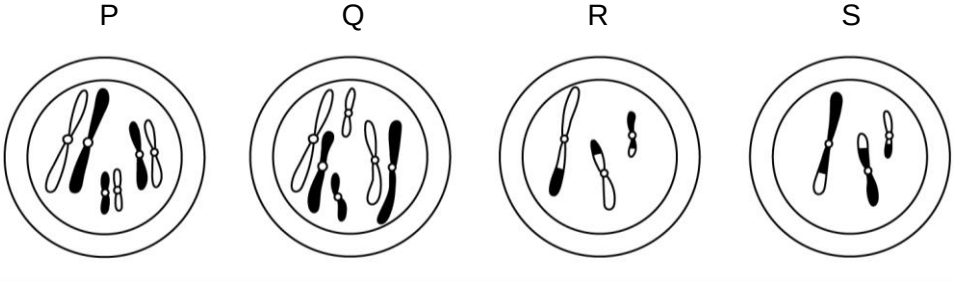
(10 marks)

##

	Skill assessed	Mark(s)
(a)	Analysing data	3
(b)	Analysing data	2
(c)	Understanding basic concepts	5

|!|BCELB0120000005|!| ★

The diagram below shows four cells P, Q, R and S. Cells Q, R and S are originated from cell P.



(a) Using the letters in the diagram, identify the cell(s) in the diagram that is/are diploid. (1 mark)

(b) What is the haploid chromosome number for cell P? (1 mark)

(c) Which cell(s), Q, R or S is/are produced by mitosis? Explain your answer. (2 marks)

(d) In which human organs do both mitotic cell division and meiotic cell division occur? (2 marks)

(e) Assuming that no crossing over occurs and cell P divides by meiotic cell division, draw all the possible daughter cells that could be produced. (4 marks)

##

(a) Cells P and Q (1)

(b) 3 (1)

(c) Cell Q (1)

because the chromosomes of cell Q and cell P are identical. (1)

(d) Testis / testes (1)

Ovary / ovaries (1)

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(e) Correct drawing (4)

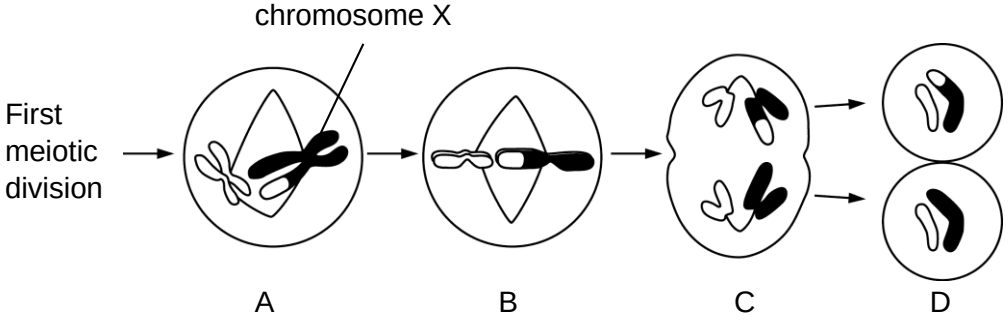


##

	Skill assessed	Mark(s)
(a)	Analysing data	1
(b)	Analysing data	1
(c)	Analysing data	2
(d)	Understanding basic concepts	2
(e)	Presenting information	4

|!|BCELA0120000006|!| ★★ **Cross topic**

The following diagram shows the stages that occur during the second meiotic division.



(a) Identify stages B and C. Give the evidence from the diagram that you used to identify the stage. (4 marks)

(b) In the above diagram, chromosome X contains a section of genetic material from the chromosome that was homologous to X. The two homologous chromosomes exchanged genetic material during the first meiotic division.

(i) Identify the stage in the first meiotic division in which the exchange of genetic material occurs. (1 mark)

(ii) What is the significance of the exchange of genetic material? (1 mark)

(c) Colchicine is a drug used to treat gout. However, colchicine has been found to be able to prevent the formation of spindle fibres during cell division.

(i) Describe how colchicine may affect the second meiotic division and hence lead to chromosome mutation in the gamete formation. (2 marks)

(ii) Describe how Down syndrome occurs as a result of chromosome mutation in the gamete formation. (1 mark)

##

(a) Stage B: Metaphase II (1)
Evidence: The non-homologous chromosomes line up along the equatorial plane of the cell. (1)
Stage C: Anaphase II (1)
Evidence: The sister chromatids are pulled towards opposite poles of the cell. (1)

(b) (i) Prophase I (1)
(ii) It produces new genetic combinations in the gametes / gives rise to genetic variation. (1)

(c) (i) Colchicine prevents the separation of sister chromatids. (1)
Therefore, the sister chromatids will move together to the same gamete, leading to an extra chromosome in the gamete. (1)
(ii) When a gamete containing an extra 21 chromosome fuses with a normal gamete, the offspring produced will have Down syndrome. (1)

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(9 marks)

##

	Skill assessed	Mark(s)
(a)	Analysing data	4
(b)(i)	Understanding basic concepts	1
(b)(ii)	Understanding basic concepts	1
(c)(i)	Applying concepts	2
(c)(ii)	Applying concepts	1

|!|BCELC0120000007|!| **Cross topic**

Simon received from his father a male rabbit and a female rabbit as birthday presents. After six months, the female rabbit gave birth to eight babies.

(a) Sexual reproduction in rabbits involves gamete formation.

(i) State the type of cell division that produces gametes. (1 mark)

_____

(ii) State the organs in which the gametes are formed in female and male rabbits. (2 marks)

_____

(b) The gametes fuse to form a zygote.

(i) Where does the fusion of gametes take place in the female rabbit? (1 mark)

_____

(ii) Describe how the zygote develops into an embryo. (1 mark)

_____

_____

(c) The baby rabbits look different from one another. List the events in cell division that cause the differences. (3 marks)

_____

_____

_____

_____

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##

(a) (i) Meiotic cell division (1)

(ii) Female: Ovary (1)

Male: Testis (1)

(b) (i) Oviduct (1)

(ii) The zygote undergoes mitotic cell division repeatedly to form an embryo. (1)

(c) Independent assortment of chromosomes that happens in the first division of meiosis (1)

Crossing over that may happen during prophase I of meiosis (1)

Random fusion of gametes that happens during fertilization (1)

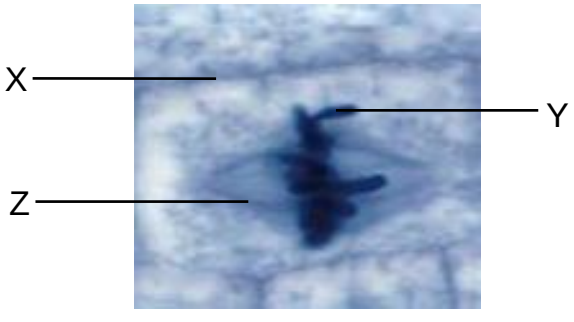
(8 marks)

##

	Skill assessed	Mark(s)
(a)(i)	Understanding basic concepts	1
(a)(ii)	Understanding basic concepts	2
(b)(i)	Understanding basic concepts	1
(b)(ii)	Understanding basic concepts	1
(c)	Applying concepts	3

|!|BCELB0120000008|!| ★

The following photomicrograph shows a cell undergoing cell division.



(a) Identify structure X. State its main component. (2 marks)

(b) State ***one*** function of structures Y and Z. (2 marks)

(c) What is the stage of cell division as shown in the photomicrograph? (1 mark)

(d) Describe how this cell will finally become two daughter cells. (4 marks)

##

(a) Cell wall (1)
Cellulose (1)

(b) Structure Y contains the genetic information which is passed from parent cells to daughter cells. (1)
Structure Z contracts and pull the chromosomes / chromatids towards opposite poles of the cells. (1)

(c) Metaphase (1)

(d) The sister chromatids of each chromosome are pulled apart towards opposite poles of the cell. (1)
A nuclear membrane forms around each set of daughter chromosomes. (1)
The chromosomes uncoil to become chromatin again. (1)
A cell plate is formed between two daughter nuclei and this divides the cell into two. (1)

(9 marks)

##

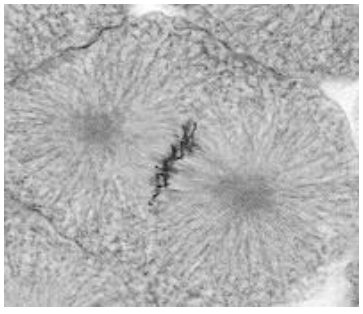
	Skill assessed	Mark(s)
(a)	Analysing data	1
	Understanding basic concepts	1
(b)	Understanding basic concepts	2
(c)	Analysing data	1
(d)	Understanding basic concepts	4

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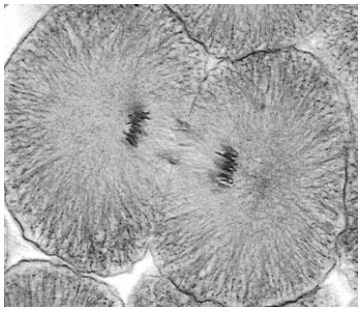
|!|BCELC0120000009|!|

Amy examined a slide showing a type of cell division in human cells. She identified four different stages of the cell division.

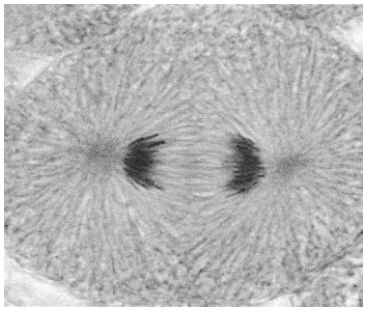
A.



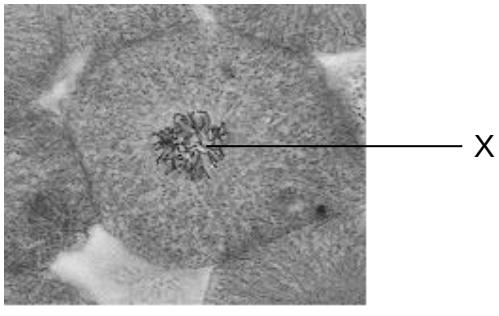
B.



C.



D.



(a) Arrange the four photomicrographs in the correct sequence. (2 marks)

(b) Identify the type of cell division. (1 mark)

(c) (i) Name structure X. (1 mark)

(ii) Give the total number of structure X in the parent cell. (1 mark)

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(iii)Briefly describe this type of cell division and show how this cell division can produces daughter cells with the number of structure X the same as that in (c)(ii). (4 marks)

##

(a) D → A → C → B (2)

(b) Mitotic cell division (1)

(c) (i) Chromosome (1)

(ii) 46 (1)

(iii)The replicated chromosomes become visible as pairs of chromatids. (1)

Then each pair of sister chromatids separates (1)

and moves to the opposite poles of the cell. (1)

The cell splits into two daughter cells, each containing one complete set of chromosome / diploid number of chromosomes. (1)

(9 marks)

##

	Skill assessed	Mark(s)
(a)	Analysing data	2
(b)	Analysing data	1
(c)(i)	Analysing data	1
(c)(ii)	Understanding basic concepts	1
(c)(iii)	Understanding basic concepts	4

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|!|BCELA0120000010|!| ★★

Figure 1 shows some cells in an onion root tip undergoing cell division and ***Figure 2*** shows the changes in the mass of DNA during the various stages of the cell cycle:

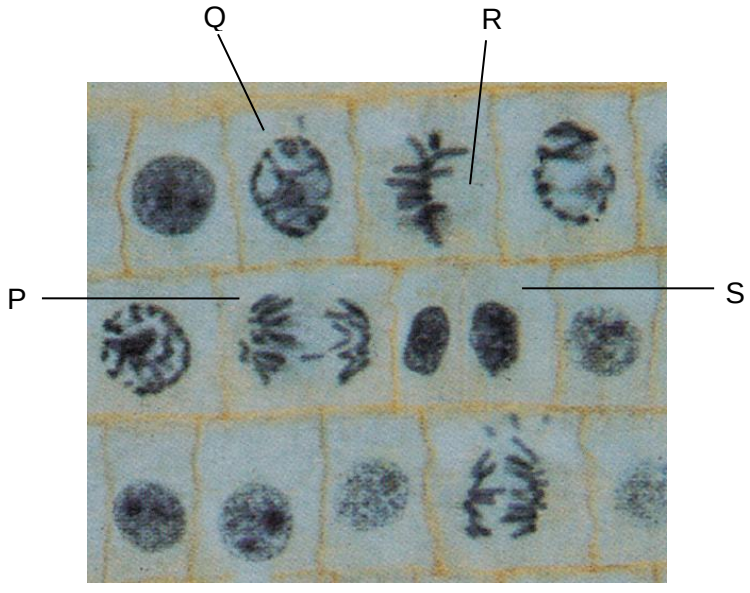


Figure 1

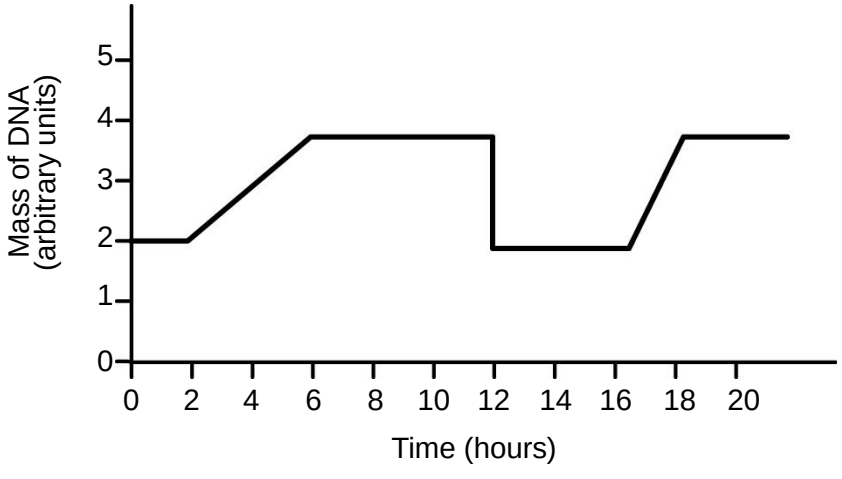


Figure 2

(a) Refer to ***Figure 1***,

(i) name the stages P and Q. (2 marks)

_____

_____

(ii) describe what happens during stage R. (2 marks)

(b) With reference to ***Figure 2***,

(i) state the time at which the stage S occurs. (1 mark)

(ii) what is the mass of DNA for the cell at stage R? (1 mark)

(iii) state ***two*** processes that occur in the cells during the period from 12 to 18 hours. (2 marks)

##

(a) (i) Stage P: Anaphase (1)
Stage Q: Prophase (1)
(ii) Chromosomes line up along the equatorial plane of the cell (1)
and attach to spindle fibres by their centromeres. (1)

(b) (i) 12 hours (1)
(ii) 4 arbitrary units (1)
(iii) Any ***two*** of the following: (1 mark each) (2)
- Proteins synthesis (1)
- Duplication of new organelles (1)
- Replication of DNA (1)
(Or other reasonable answers)

(8 marks)

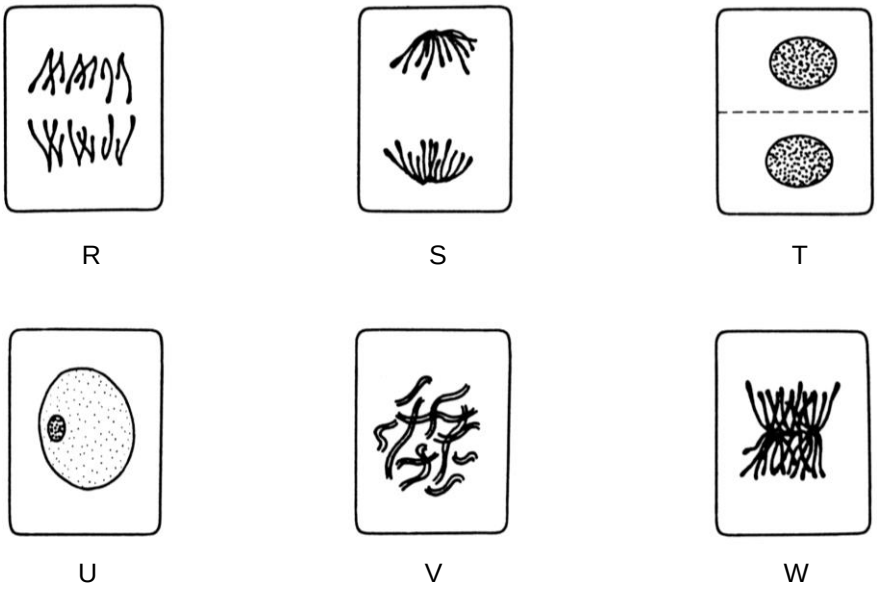
##

	Skill assessed	Mark(s)
(a)(i)	Analysing data	2
(a)(ii)	Understanding basic concepts	2
(b)(i)	Analysing data	1
(b)(ii)	Analysing data	1
(b)(iii)	Applying concepts	2

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|!|BCELC0120000011|!|

The following diagrams show some stages of cell division.



(a) Name this type of cell division. (1 mark)

(b) Arrange the above stages of cell division in the correct order. (2 marks)

(c) Where does this type of cell division take place in a plant? (1 mark)

(d) Describe what happens during stage S. (2 marks)

(e) State ***two*** significance of this type of cell division. (2 marks)

##

(a) Mitotic cell division (1)

(b) U → V → W → R → S → T (2)

(c) Root tip / stem tip / meristem (1)

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(d) In stage S, the spindle fibres which attach to the chromosomes contract. (1)
Sister chromatids of each chromosome separate and move to the opposite poles of cells. (1)

(e) Mitotic cell division provides new cells for growth and repair. (1)
Some organisms reproduce asexually by mitotic cell division. (1)

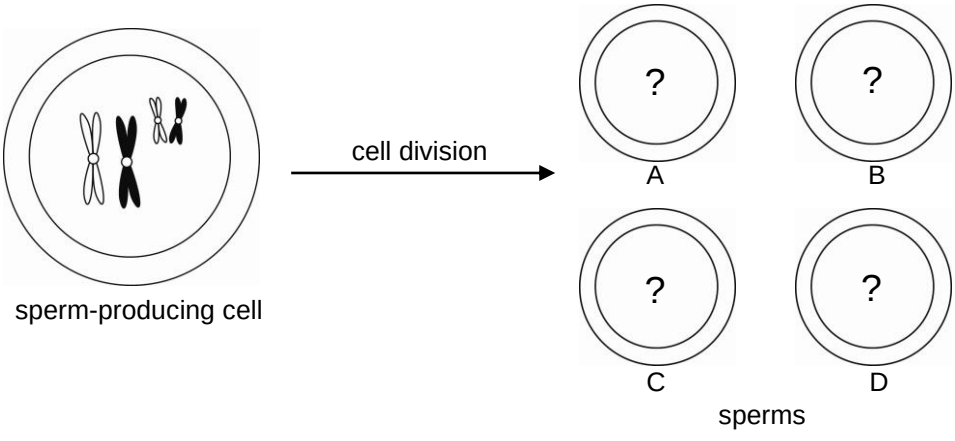
(8 marks)

##

	Skill assessed	Mark(s)
(a)	Analysing data	1
(b)	Understanding basic concepts	2
(c)	Understanding basic concepts	1
(d)	Understanding basic concepts	2
(e)	Understanding basic concepts	2

|!|BCELB0120000012|!| ★ **Cross topic**

The following diagram shows a type of cell division. (Only two pairs of chromosomes are shown.)



(a) Which type of cell division is carried out by the sperm-producing cell? What is the significance of this type of cell division? (5 marks)

(b) Draw diagrams to show the different chromosome contents in sperms A, B, C and D. (2 marks)

(c) In which organ does this type of cell division take place in a mammal and a flowering plant? (2 marks)

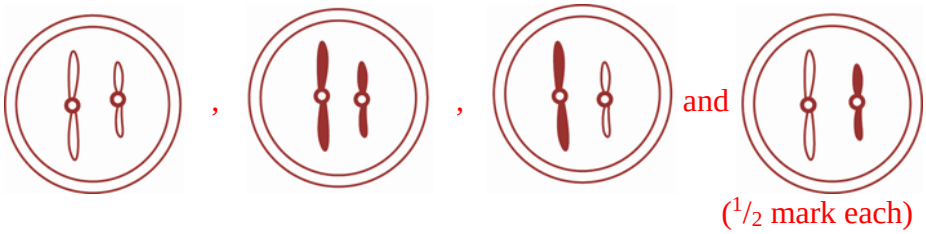
##

(a) Meiotic cell division (1)

Significance:

1. It ensures that the gametes produced are haploid (n) so that when they fuse during fertilization, the diploid (2n) number of chromosomes is restored. (2)
2. The independent assortment and crossing over of homologous chromosomes takes place during meiosis, resulting in genetic variation among offspring. (2)

(b)



($^1/_2$ mark each) (2)

(c) In mammals: Testis or ovary (1)

In flowering plants: Anther or ovule (1)

(9 marks)

##

	Skill assessed	Mark(s)
(a)	Analysing data	1
	Understanding basic concepts	4
(b)	Presenting information	2
(c)	Understanding basic concepts	2

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